SETTING UP A SECURE WEB SERVER ON UBUNTU

Test 6. WEB SERVER SECURITY – Exercise 6

Valentina Hatos – 26.12.2025

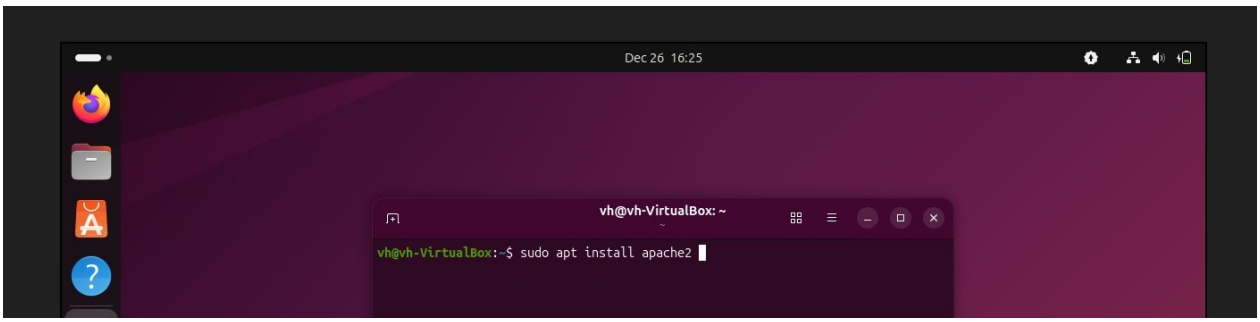
As a preliminary step, I installed Ubuntu Server on an Oracle VirtualBox virtual machine to meet the requirements of the test.

Step 1: Update the system

sudo apt update && sudo apt upgrade -y

Step 2: Install Apache

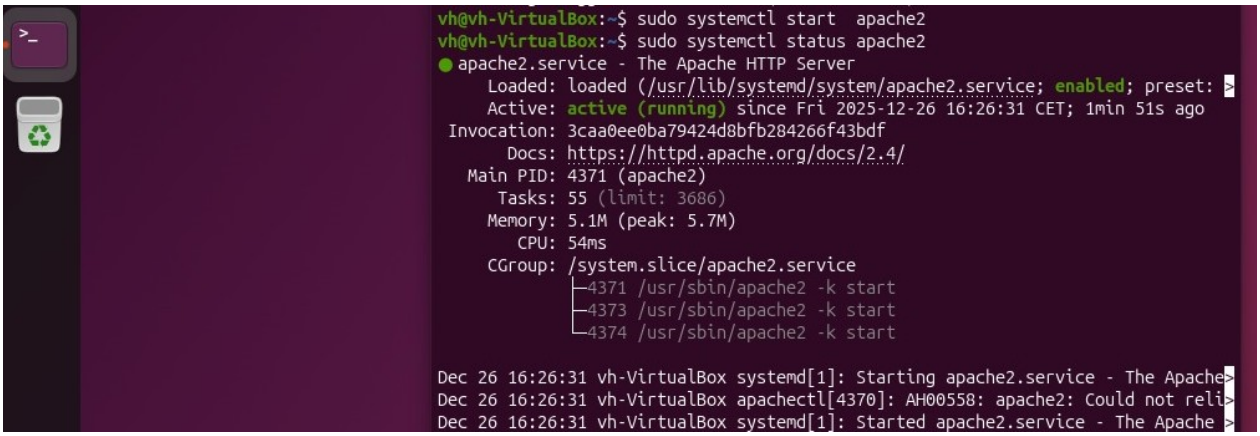
sudo apt install apache2



Step 3: Check that it is active

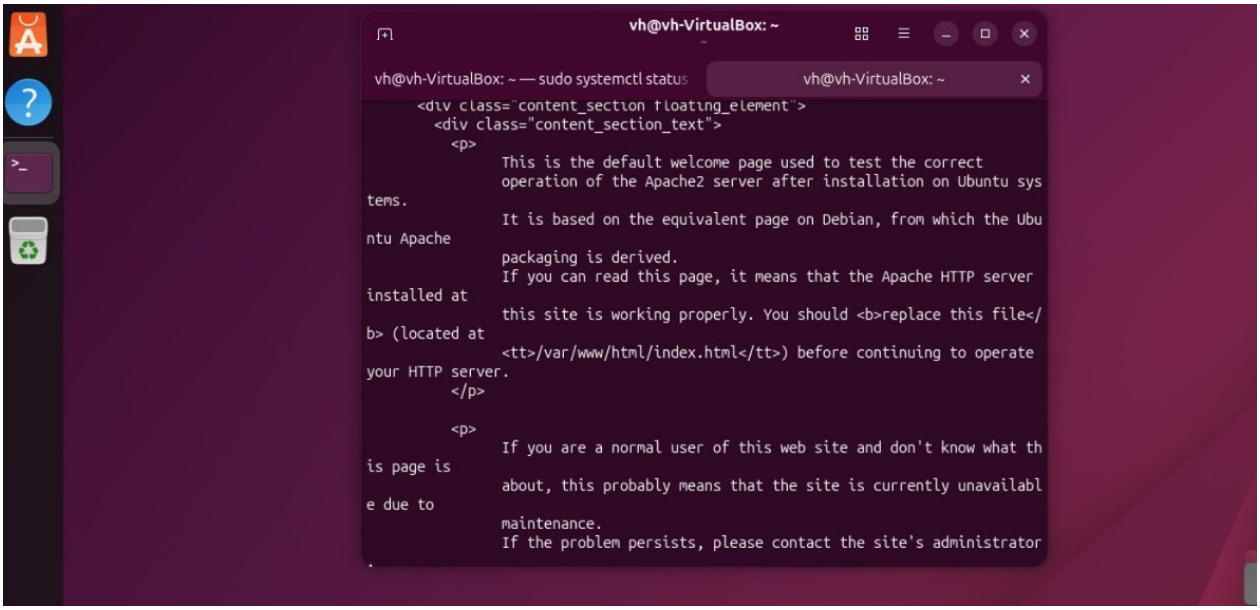
sudo systemctl start apache2

sudo systemctl status apache2



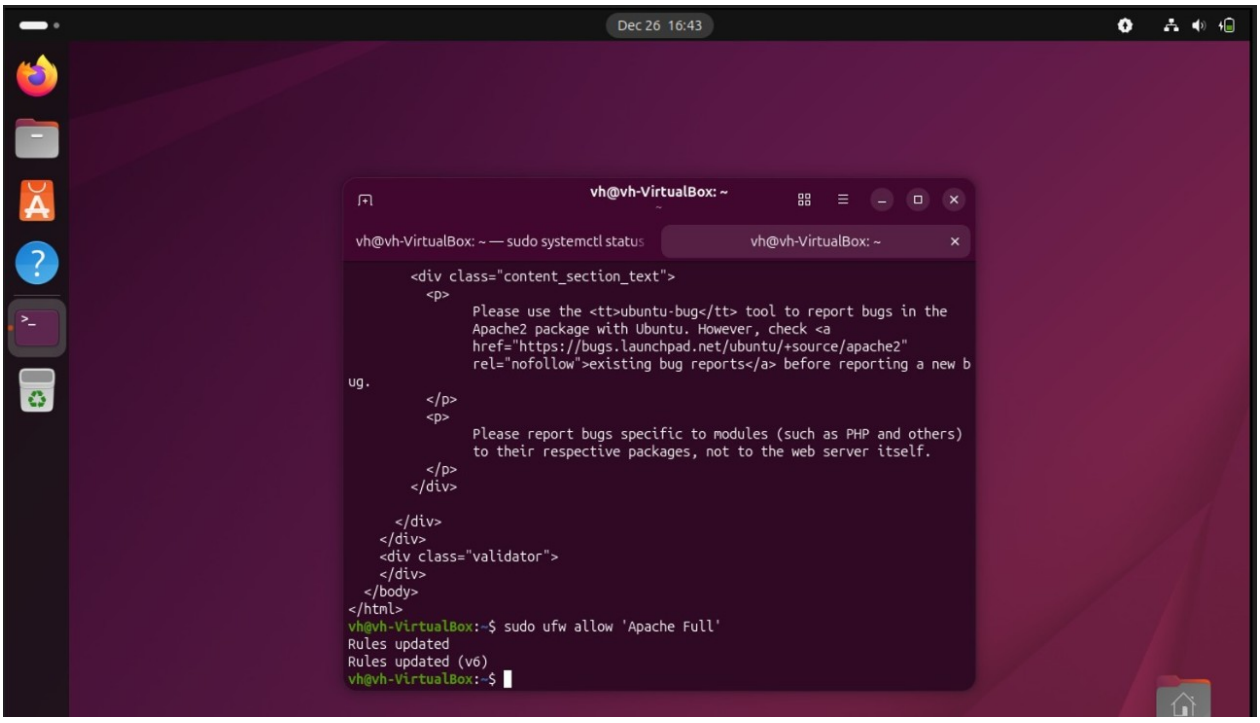
Step 4: Test that it works

curl http://localhost



Step 5: Configure the firewall (to allow HTTP and HTTPS traffic)

sudo ufw allow 'Apache Full'



Step 6: Directory for hosting sites

- by default, Apache uses:

/var/www/html

Step 7: Enable HTTPS with SSL

- install the SSL modules:

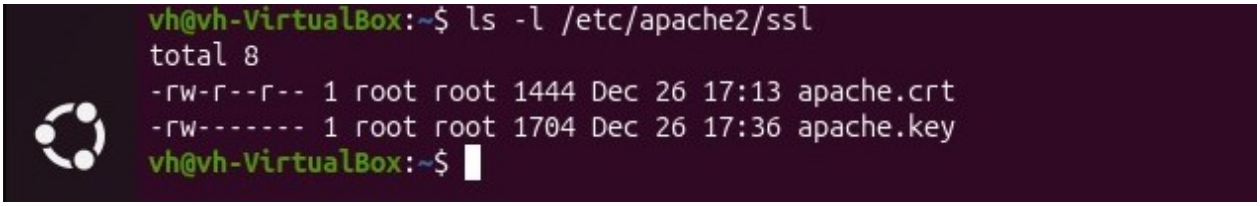
sudo apt install openssl apache2 ssl-cert -y

sudo a2enmod ssl

- Create a self-signed certificate:

sudo mkdir /etc/apache2/ssl

```
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
    -keyout /etc/apache2/ssl/apache.key \
    -out /etc/apache2/ssl/apache.crt
```



- Configure the SSL Virtual Host:

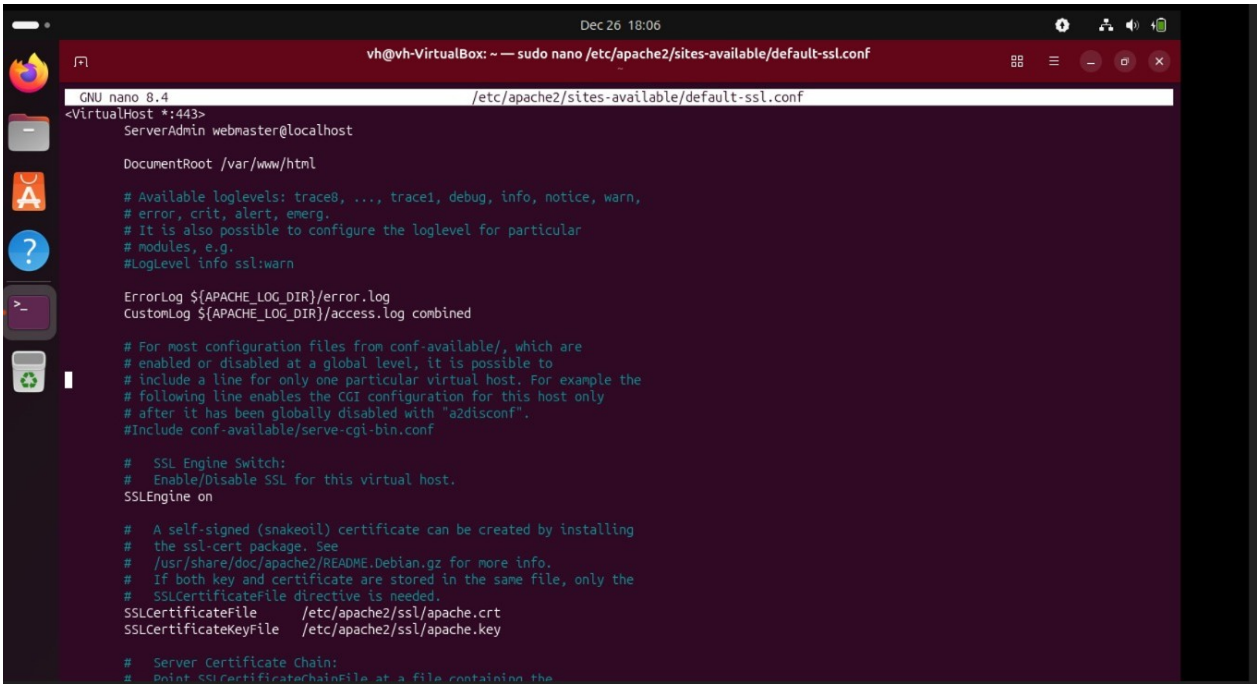
sudo nano /etc/apache2/sites-available/default-ssl.conf

Add:

SSLEngine on

SSLCertificateFile /etc/apache2/ssl/apache.crt

SSLCertificateKeyFile /etc/apache2/ssl/apache.key



Enable the SSL site:

sudo a2ensite default-ssl.conf

sudo systemctl reload apache2

Final result:

Secure Ubuntu Server, with Apache installed and configured with HTTPS to protect communications.

Recommended security best practices:

- Always use HTTPS (valid certificates in production, not self-signed);
- Update the system and Apache regularly;
- Set correct permissions on /var/www/html;
- Disable unnecessary Apache modules;
- Install ModSecurity and ModEvasive for protection against XSS and DoS;
- Configure fail2ban to block brute-force attempts;
- Make regular backups;
- Monitor the server with tools such as Nagios.